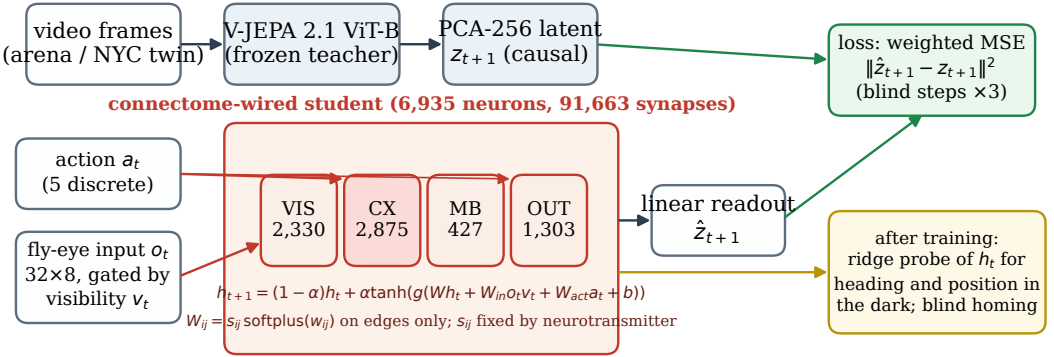
video frames
(arena / NYC twin)
V-JEPA 2.1 ViT-B
(frozen teacher)
PCA-256 latent
$z_{t+1}$ (causal)
loss: weighted MSE
$\|\hat{z}_{t+1} - z_{t+1}\|^2$
(blind steps ×3)
connectome-wired student (6,935 neurons, 91,663 synapses)
action $a_t$
(5 discrete)
fly-eye input $o_t$
32×8, gated by
visibility $v_t$
VIS
2,330
CX
2,875
MB
427
OUT
1,303
linear readout
$\hat{z}_{t+1}$
after training:
ridge probe of $h_t$ for
heading and position in
the dark; blind homing
$h_{t+1} = (1-\alpha)h_t + \alpha\tanh(g(Wh_t + W_{in}o_t v_t + W_{act}a_t + b))$
$W_{ij} = s_{ij}\,\mathrm{softplus}(w_{ij})$ on edges only; $s_{ij}$ fixed by neurotransmitter
blackout: $v_t = 0$ on a random window during training; test windows 8:16, 8:24, 4:24 (actions still given)